Computational Mechanics 33 (2003) 52–60 © Springer-Verlag 2003
DOI 10.1007/s00466-003-0499-z

Refined triangular discrete Mindlin flat shell elements

G. Zengjie, C. Wanji

Abstract In this paper flat shell elements are formed by the assemblage of discrete Mindlin plate elements RDKTM and either the constant strain membrane element CST or the Allman's membrane element with drilling degrees of freedom LST. The element RDKTM is a robust Mindlin plate element, which can perform uniformly thick and thin plate bending analysis. It also passes the patch test for thin plate bending, and its convergence for very thin plates can be ensured theoretically. The singularity of the stiffness matrix and membrane locking are studied for the present elements. Numerical examples are presented to show that the present models indeed possess properties of simple formulations, high accuracy for thin and thick shells, and it is free from shear locking for thin plate/shell analysis.

Keywords Mindlin plate theory, Shear locking, Drilling degrees of freedom, Displacement functions of a Timoshenko beam

1 Introduction

In the formulations of shell elements there are three distinct choices: (1) The flat shell element, that is obtained by combining a membrane element for plane elasticity and a bending element for flat plates; (2) The curved shell element, that is formulated according to shell theory; and (3) The degenerated shell element, that is formed based on the three dimensional solid theory. Among those formulations, the flat shell element is considered simplest and avoids complex shell equations. Therefore, the application of flat shell elements is widespread in engineering.

Triangular flat shell elements may be used to represent an arbitrarily shaped shell as a faceted surface for curved shell structures. In this case, convergence to the exact geometry is obtained when the mesh is refined. Usually, triangular flat shell elements consist of triangular constant strain membrane elements and triangular bending plate elements. To study this type of shell element, the first step is the choice of an applicable plate element and an applicable membrane element. However, shear locking has been a major problem for the Mindlin plate element. It is well known that for Mindlin plates that only C^0 continuity is required and that the difficulties of C^1 continuity for thin plate elements can be avoided. Moreover both thin plate and thick plate analysis can be integrated in the element model. Initially, Mindlin plate elements use strain-displacement relations to obtain bending and transverse shear strain. In this case, bending energy is written in terms of nodal rotations only, whereas shear strain energy is given in terms of both nodal rotations and deflections. When the plate becomes thin, transverse shear effects are reduced and nodal deflections become associated only with the vanishing shear energy. This is a difficult situation to uphold and shear locking is soon observed. In order to avoid this, reduced integration [1, 2] and selective integration [3, 4] technique are widely used. In the 3-noded triangular element, a single Gauss point integration is used for calculating the shear strain energy. However, it is found that such elements have low accuracy, and furthermore, they very often cannot pass the patch test for thin plates. In the re-constituting shear strain technique, the shear-strain approximation along with the edge-projection (or discrete Mindlin technique) has been used to derive 3-noded triangular Mindlin plate elements. A number of efficient 9-DOF triangular Mindlin plate elements based on the discrete constraint and the equilibrium conditions can be produced [5–7]. These elements can converge towards the discrete Kirchhoff plate bending elements when the plate thickness is very small.

Recently, a refined nonconforming element method has been developed by Chen and Cheung. Based on the Mindlin plate theory, a new triangular plate element, RDKTM, with the re-constitution of the shear strain has been established [8, 9]. Using this method and the displacement function of the Timoshenko's beam, formulation of new triangular thin/thick plate elements can be derived. It can be shown that the element RDKTM indeed possesses properties of high accuracy for thin and thick plates, is capable of passing the patch test required for Kirchhoff thin plate elements, and does not exhibit extra zero energy modes. In addition, RDKTM is free of locking for very thin plate analysis (where the thickness/span ratio is less than 10^{-30}) and its convergence can be ensured theoretically.

It is well known that when all elements connected to a node are coplanar, zero stiffness appears in the direction normal to the plane rotation (θ_z), This causes the assembly stiffness matrix at a global level to be singular [10]. In order to avoid this singularity, Zienkiewicz [10] has suggested the addition of an 3×3 artificial rotational stiffness matrix to all elements in the assembly. This matrix also

Received: 18 January 2003 / Accepted: 5 August 2003
Published online: 20 November 2003

G. Zengjie, C. Wanji (✉)
State Key Laboratory of Structural Analysis for Industrial Equipments, Dalian University of Technology, Dalian, P. R. China

appears in a more recent formulation [11]. This problem may be avoided by using rotational fictitious stiffness or by introducing drilling degrees of freedom in the formulation of the membrane element. Earlier attempts to develop membrane elements with drilling degrees of freedom were not successful [12]. An important advance was made by Allman [13] who proposed a 9-DOF compatible triangular element with quadratic varying displacement fields. This element is known to have a spurious zero energy mode, while its vertex rotations are not true rotations in the sense of plane elasticity. Another problem with this method is that triangular elements with drilling degrees of freedom encounter poor response for membrane action [14–16], especially in spherical shells and twisted beam structures [17]. Recently, an approach for adding true drilling rotations to the triangular thin flat shell element was studied by Provias and Kattis [18]. Triangular, discrete Reissner-Mindlin plate and shell elements were studied by Sydenstricker and Landau [19], however the spherical shell and twisted beam problem are not considered.

In this paper, refined 15/18-DOF triangular discrete Mindlin flat shell elements are formed by the assembly of triangular discrete Mindlin plate elements, RDKTM, and triangular constant strain membrane elements, CST/Allman's triangular membrane element with drilling rotations LST.

2
Bending strain mode of element RDKTM [9]

2.1
Timoshenko's beam function

The Timoshenko's beam function can be used to derive more efficient Mindlin plate elements, whichcan be expressed as follows,

$$\begin{aligned} w &= (L_i + \mu_e L_i L_j (L_i - L_j))w_i \\ &\quad + (L_i L_j + \mu_e L_i L_j (L_i - L_j))L/2\theta_i \\ &\quad + (L_j + \mu_e L_i L_j (L_j - L_i))w_j \\ &\quad + (-L_i L_j + \mu_e L_i L_j (L_i - L_j))L/2\theta_j \\ \theta &= -(6L_i L_j/L)\mu_e w_i + L_i(1 - 3\mu_e L_j)\theta_i \\ &\quad + (6L_i L_j/L)\mu_e w_j + L_j(1 - 3\mu_e L_i)\theta_j \end{aligned} \tag{1}$$

where $L_i = 1 - x/L, L_j = x/L, \mu_e = 1/(1 + 12\lambda_e), \lambda_e = h^2/5(1 - v^2)L^2, L$ is the length of the strip, h is the height of the strip, v is Poisson's ratio.

2.2
Displacement function of the element RDKTM [9]

By eliminating the surplus parameters at the mid-side nodes as the procedure of the element DKT, the explicit expression of the rotations θ_x and θ_y of the element RDKTM for describing bending strains of the element can be obtained as follows,

$$\begin{Bmatrix} \theta_x \\ \theta_y \end{Bmatrix} = \tilde{\mathbf{N}}\mathbf{q} \tag{2}$$

where,

$\tilde{\mathbf{N}} = [\tilde{\mathbf{N}}_1 \tilde{\mathbf{N}}_2 \tilde{\mathbf{N}}_3]$ and

$$\tilde{\mathbf{N}}_i = \begin{bmatrix} P_i & P_{xi} & P_{yi} \\ Q_i & Q_{xi} & Q_{yi} \end{bmatrix} \quad (i = 1, 2, 3) \ . \tag{3}$$

For shape function $\tilde{\mathbf{N}}_1$,

$$\begin{aligned} P_1 &= 1.5_1(\mu_{12} m_{12} N_4/S_{12} - \mu_{31} m_{31} N_6/S_{31}) \\ P_{x1} &= -0.75(\mu_{12} m_{12}^2 N_4 + \mu_{31} m_{31}^2 N_6) + L_1 \\ P_{y1} &= 0.75(\mu_{12} l_{12} m_{12} N_4 + \mu_{31} l_{31} m_{31} N_6) \\ Q_1 &= 1.5(-\mu_{12} l_{12} N_4/S_{12} + \mu_{31} l_{31} N_6/S_{31}) \\ Q_{x1} &= 0.75(\mu_{12} l_{12} m_{12} N_4 + \mu_{31} l_{31} m_{31} N_6) \\ Q_{y1} &= -0.75(\mu_{12} l_{12}^2 N_4 + \mu_{31} l_{31}^2 N_6) + L_1 \end{aligned} \tag{4}$$

where, $\mu_{ij} = 1/(1 + 12\lambda_{ij})$, $\lambda_{ij} = t^2/5S_{ij}^2(1 - v)$, S_{ij} is the $i - j$ boundary length,
$\mathbf{q} = [\, w_1 \quad \theta_{x1} \quad \theta_{y1} \quad w_2 \quad \theta_{x2} \quad \theta_{y2} \quad w_3 \quad \theta_{x3} \quad \theta_{y3} \,]^{\mathrm{T}}$, and N_i is the shape function of the 6-node triangular element in area coordinates (L_i), and

$$\begin{aligned} N_i &= (2L_i - 1)L_i \quad (i = 1, 2, 3) \\ N_k &= 4L_i L_j \quad (k = 4, 5, 6; \ \ ij = 12, 23, 31) \end{aligned} \tag{5}$$

Other shape functions $\tilde{\mathbf{N}}_i$ $(i = 2, 3)$ can be obtained by cyclic permutation.

It is obvious that element displacements in Eq. (4) will become the displacements of the discrete Kirchhoff plate element DKT, because of $\lambda_{ij} \to 0$ when $t/S_{ij} \to 0$.

2.3
The bending strain and the bending part of the stiffness matrix of element RDKTM

The bending strain of the element can be written as

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \frac{\partial \theta_x}{\partial x} \\ \frac{\partial \theta_y}{\partial y} \\ \frac{\partial \theta_x}{\partial y} + \frac{\partial \theta_y}{\partial x} \end{bmatrix} \tag{6}$$

The bending strain $\boldsymbol{\epsilon}$ can be expressed as

$$\boldsymbol{\varepsilon} = \mathbf{B}_b \mathbf{q} \tag{7}$$

The part of bending of the element stiffness matrix can be written as follows,

$$\mathbf{K}_b^e = \int_{V_e} \mathbf{B}_b^T \mathbf{D}_b \mathbf{B}_b \, \mathrm{d}x \, \mathrm{d}y \tag{8}$$

where $\mathbf{D}_b$ is the elasticity matrix, and

$$\mathbf{D}_b = \bar{D}_b \begin{bmatrix} 1 & v & 0 \\ v & 1 & 0 \\ 0 & 0 & (1 - v)/2 \end{bmatrix}, \ \bar{D}_b = \tfrac{Et^3}{12(1 - v^2)} .$$

3
Shear strain mode of element RDKTM [9]

The shear strain can be expressed as follows,

$$\gamma = \begin{Bmatrix} \gamma_x \\ \gamma_y \end{Bmatrix} = \begin{Bmatrix} \theta_x - \frac{\partial w}{\partial x} \\ \theta_y - \frac{\partial w}{\partial y} \end{Bmatrix} \tag{9}$$

$$\begin{Bmatrix}\gamma_x\\ \gamma_y\end{Bmatrix}=\begin{bmatrix}L_1 & 0 & L_2 & 0 & L_3 & 0\\ 0 & L_1 & 0 & L_2 & 0 & L_3\end{bmatrix}\begin{Bmatrix}\gamma_{x1}\\ \gamma_{y1}\\ \gamma_{x2}\\ \gamma_{y2}\\ \gamma_{x3}\\ \gamma_{y3}\end{Bmatrix} \tag{10}$$

Timoshenko's beam function can be used to define the deflection on the element boundary. The interpolation of the displacements $\tilde{w}_s$ on the $i-j$ boundary (see Fig. 1(b)) by using the parameters of the i and j nodes which are w_i, θ_{si} and w_j, θ_{sj} can be given as

$$\begin{aligned}\tilde{w}_s &= (L_i+\mu_{ij}L_iL_j(L_i-L_j))w_i\\ &\quad+(L_iL_j+\mu_{ij}L_iL_j(L_i-L_j))S_{ij}/2\theta_{si}\\ &\quad+(L_j+\mu_{ij}L_iL_j(L_j-L_i))w_j\\ &\quad+(-L_iL_j+\mu_{ij}L_iL_j(L_i-L_j))S_{ij}/2\theta_{sj}\end{aligned} \tag{11}$$

and thus giving,

$$\begin{aligned}\frac{\partial\tilde{w}_s}{\partial s} &= \frac{1}{S_{ij}}(-1+\mu_{ij}(1-6L_iL_j))w_i\\ &\quad+0.5(L_i-L_j+\mu_{ij}(1-6L_iL_j))\theta_{Si}\\ &\quad+\frac{1}{S_{ij}}(1-\mu_{ij}(1-6L_iL_j))w_j\\ &\quad+0.5(-L_i+L_j+\mu_{ij}(1-6L_iL_j))\theta_{Sj}\end{aligned} \tag{12}$$

It is clear that the displacements $\tilde{\theta}_s-\partial\tilde{w}_s/\partial S$ on the boundary will always be constant because of

$$\begin{aligned}\tilde{\theta}_s-\frac{\partial\tilde{w}_s}{\partial S} &= (1-\mu_{ij})w_i/S_{ij}+0.5(1-\mu_{ij})\theta_{Si}\\ &\quad-(1-\mu_{ij})w_j/S_{ij}+0.5(1-\mu_{ij})\theta_{Sj}\end{aligned} \tag{13}$$

Therefore the shear strains at node i can be expressed by the constant shear strains, leading to the following expressions for node 1 (see Fig. 1(a)),

$$\begin{Bmatrix}\gamma_{s4}\\ \gamma_{s6}\end{Bmatrix}=\begin{bmatrix}-m_{12} & l_{12}\\ -m_{31} & l_{31}\end{bmatrix}\begin{Bmatrix}\gamma_{x1}\\ \gamma_{y1}\end{Bmatrix} \tag{14}$$

and

$$\begin{Bmatrix}\gamma_{x1}\\ \gamma_{y1}\end{Bmatrix}=\frac{1}{l_{12}m_{31}-l_{31}m_{12}}\begin{bmatrix}l_{31} & -l_{12}\\ m_{31} & -m_{12}\end{bmatrix}\begin{Bmatrix}\gamma_{s4}\\ \gamma_{s6}\end{Bmatrix}. \tag{15}$$

Similarly, the remaining nodal shear strains γ_{xi} and $\gamma_{yi}(i=2,3)$ can be obtained by cyclic permutation.

Finally the shear strains can be obtained as follows,

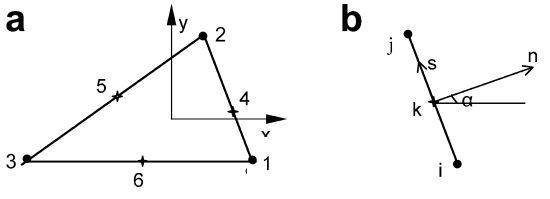


Fig. 1. Triangular discrete Mindlin plate element

$$\begin{Bmatrix}\gamma_x\\ \gamma_y\end{Bmatrix}=\tilde{\mathbf{N}}\begin{Bmatrix}\gamma_{s4}\\ \gamma_{s5}\\ \gamma_{s6}\end{Bmatrix}$$

and

$$\tilde{\mathbf{N}}=\begin{bmatrix}\bar{l}_3L_1-\tilde{l}_2L_2 & \tilde{l}_1L_2-\hat{l}_3L_3 & \hat{l}_2L_3-\bar{l}_1L_1\\ \bar{m}_3L_1-\tilde{m}_2L_2 & \tilde{m}_1L_2-\hat{m}_3L_3 & \hat{m}_2L_3-\bar{m}_1L_1\end{bmatrix} \tag{16}$$

where
$\bar{l}_i=l_i/A_1$, $\tilde{l}_i=l_i/A_2$, $\hat{l}_i=l_i/A_3$, $\bar{m}_i=m_i/A_1$, $\tilde{m}_i=m_i/A_2$, $\hat{m}_i=m_i/A_3(i=12,23,31)$, $A_1=-l_{31}m_{12}+l_{12}m_{31}$, $A_2=-l_{12}m_{23}+l_{23}m_{12}$, $A_3=-l_{23}m_{31}+l_{31}m_{23}$.
$\gamma_{sk}(k=4,5,6)$ are the natural shear strains at mid-side nodes 4, 5, 6 of the element (see Fig.1 (a)). This means the shear strains can be expressed as follows,

$$\begin{Bmatrix}\gamma_{s4}\\ \gamma_{s5}\\ \gamma_{s6}\end{Bmatrix}=\begin{Bmatrix}\theta_{s4}-\frac{\partial w_{s4}}{\partial S}\\ \theta_{s5}-\frac{\partial w_{s5}}{\partial S}\\ \theta_{s6}-\frac{\partial w_{s6}}{\partial S}\end{Bmatrix} \tag{17}$$

Substituting $L_i=L_j=0.5$ into Eq. (13), the following is determined for node k (see Fig. 1(b))

$$\begin{aligned}\theta_{sk}-\frac{\partial w_{sk}}{\partial S} &= (1-\mu_{ij})w_i/S_{ij}+0.5(1-\mu_{ij})\theta_{Si}\\ &\quad-(1-\mu_{ij})w_j/S_{ij}+0.5(1-\mu_{ij})\theta_{Sj}\end{aligned} \tag{18}$$

It is obvious that $\gamma_x,\gamma_y\to 0$ when $(t/S_{ij})^2\to 0, \mu_{ij}=1$. The element RDTMK is free of shear locking for thin plate analysis and its convergence can be ensured theoretically. Actually, the solution of element RDKTM is the same as element DKT when $(t/S_{ij})<10^{-10}$.

From $\theta_s=\begin{bmatrix}-m_{ij} & l_{ij}\end{bmatrix}\begin{Bmatrix}\theta_x\\ \theta_y\end{Bmatrix}$, it can be found that,

$$\theta_{sk}-\frac{\partial w_{sk}}{\partial S}=\tilde{\mathbf{T}}_k\begin{Bmatrix}w_i\\ \theta_{xi}\\ \theta_{yi}\\ w_j\\ \theta_{xj}\\ \theta_{yj}\end{Bmatrix} \tag{19}$$

Finally, the mid-side node parameters $(\theta_{sk}-\partial w_{sk}/\partial s)$ of the element RDKTM can be expressed as

$$\theta_{sk}-\frac{\partial w_{sk}}{\partial s}=\mathbf{B}_{sij}\mathbf{q}\quad(k=4,5,6;\ \ ij=12,23,31) \tag{20}$$

The shear strains of the element RDKTM can be obtained as follows:

$$\gamma=\begin{Bmatrix}\theta_x-\frac{\partial w}{\partial x}\\ \theta_y-\frac{\partial w}{\partial y}\end{Bmatrix}=\tilde{\mathbf{N}}\begin{Bmatrix}\theta_{s4}-\frac{\partial w_{s4}}{\partial S}\\ \theta_{s5}-\frac{\partial w_{s5}}{\partial S}\\ \theta_{s6}-\frac{\partial w_{s6}}{\partial S}\end{Bmatrix}=\tilde{\mathbf{N}}\tilde{\mathbf{B}}_s\mathbf{q}=\mathbf{B}_s\mathbf{q} \tag{21}$$

where

$$\tilde{B}_s=\begin{bmatrix}B_{s12}\\ B_{s23}\\ B_{s31}\end{bmatrix} \tag{22}$$

It should be noted that according to the Timoshenko beam function, the shear strain γ_s on the element boundary is constant and can be denoted as γ_{s4}, γ_{s5} and γ_{s6}, therefore γ_{xi} and γ_{yi} can be denoted at the mi-side node i. The element shear strains, γ_x and γ_y can be found by using γ_{s4}, γ_{s5} and γ_{s6}. The resulting shear strains γ_x and γ_y of the element are linearly distributed as their presumed values are distributed linearly. The patch test is presented to indicate that γ_x and γ_y can exactly express zero constant shear strain, and are free of shear locking.

The shear part of the stiffness matrix of the element RDKTM can be written as follows,

$$\mathbf{K}_s^e = \int_{V_e} \mathbf{B}_s^{\mathrm{T}} \mathbf{D}_s \mathbf{B}_s \, \mathrm{d}x \, \mathrm{d}y \tag{23}$$

where $\mathbf{D}_s$ is the elasticity matrix, and $\mathbf{D}_s = \bar{D}_s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\bar{D}_s = 5Et/12(1+\nu)$.

The stiffness matrix of element RDKTM can be written as follows,

$$\mathbf{K}_p^e = \mathbf{K}_b^e + \mathbf{K}_s^e \tag{24}$$

4
Triangular flat shell elements RDKTM15

The stiffness matrix of a flat shell element in a local coordinate system is obtained by simple assembling of a membrane element and plate element stiffness matrix,

$$\mathbf{K}^e = \mathbf{K}_m^e + \mathbf{K}_p^e \tag{25}$$

In this case, the nodal displacement vector of the shell element is organized as

$$\mathbf{a} = \mathbf{a}_m + \mathbf{a}_p \tag{26}$$

where $\mathbf{a}_m$ is the membrane nodal displacement vector, and $\mathbf{a}_m = [\,u \quad v\,]^{\mathrm{T}}$; $\mathbf{a}_p$ is the plate nodal displacement vector, and $\mathbf{a}_p = [\,w \quad \theta_x \quad \theta_y\,]^{\mathrm{T}}$.

The above formulation of the flat shell elements will lead to singularities if all elements are coplanar. Such singularities can be suppressed by adding a stabilization stiffness matrix $\mathbf{K}_\theta^e$ to each element stiffness matrix in the local coordinates $\mathbf{K}^e$ [10].

$$\mathbf{K}^e = \mathbf{K}_m^e + \mathbf{K}_p^e + \mathbf{K}_\theta^e \tag{27}$$

The stabilization matrix is given by

$$\begin{Bmatrix} M_{z1} \\ M_{z2} \\ M_{z3} \end{Bmatrix} = \alpha EA \begin{bmatrix} 1 & -0.5 & -0.5 \\ -0.5 & 1 & -0.5 \\ -0.5 & -0.5 & 1 \end{bmatrix} \begin{Bmatrix} \theta_{z1} \\ \theta_{z2} \\ \theta_{z3} \end{Bmatrix} \tag{28}$$

$$K_\theta^e = \alpha EA \begin{bmatrix} 1 & -0.5 & -0.5 \\ -0.5 & 1 & -0.5 \\ -0.5 & -0.5 & 1 \end{bmatrix} \tag{29}$$

where $\theta_{zi} (i = 1, 2, 3)$ is the drilling DOF at the node of the flat element, A is the area of the element, E is the elastic modulus, α is a small constant as $\alpha = t \times 10^{-4}$, and t is thickness of the element. Equation (28) is a penalty function and provides a fictitious stiffness for θ_z.

5
18-DOF triangular flat shell elements with fictitious drilling rotation RDKTM18 (P)

The fictitious drilling DOF is adopted to increase the order of the displacement function of the membrane element. The displacements u and v with drilling DOF have been given by Allman [13] as follows,

$$\begin{aligned} u &= \sum_{i=1}^{3} L_i u_i + p_1 \theta_{z1} + p_2 \theta_{z2} + p_3 \theta_{z3} \\ v &= \sum_{i=1}^{3} L_i v_i + q_1 \theta_{z1} + q_2 \theta_{z2} + q_3 \theta_{z3} \end{aligned} \tag{30}$$

where $L_i (i = 1, 2, 3)$ are the area coordinates of the element, u_i, v_i are the nodal displacement parameters, and θ_{zi} are the nodal rotations about the z-axis in the local coordinates of the plate element. θ_{zi} are so-called pseudo drilling degrees of freedom. The p_i, $q_i (i = 1, 2, 3)$, using $i = 1$ as an example, is expressed as follows

$$\begin{aligned} p_1 &= (b_3 L_2 - b_2 L_3) L_1 / 2 \\ q_1 &= (c_3 L_2 - c_2 L_3) L_1 / 2 \end{aligned} \tag{31}$$

and

$$\begin{aligned} b_1 &= y_2 - y_3 \\ c_1 &= x_3 - x_2 \end{aligned} \tag{32}$$

By cycling the parameters $1 \to 2 \to 3 \to 1$ the other parameters, p_i, q_i, b_i, c_i $(i = 2, 3)$ can be obtained.

6
18-DOF triangular flat shell elements with true drilling rotation RDKTM18(T)

Allman's rotation is not identical with the rotation variable defined in the theory of elasticity, and it may have an effect on the precision by showing over-stiffness. Moreover, when θ_{zi} takes the same non-zero values, extra zero energy modes are produced. To solve this problem, the rotation equations of the theory of elasticity are introduced as constraint conditions and the least square method is applied in the functional so that rotation equations are satisfied by assigning weights and averages.

For 18-DOF triangular flat shell elements with drilling rotations, the functional for an element with true drilling rotation can be written as

$$\Pi^e = \Pi_m^e + \Pi_p^e + \int_{V_e} \beta \frac{Et}{2(1-\nu^2)} (\theta_z - \bar{L}u)^{\mathrm{T}} (\theta_z - \bar{L}u) \, \mathrm{d}V \tag{33}$$

where Π_m^e is the membrane strain energy of element; Π_p^e is the bending strain energy of element; ν is Poisson's ratio; β is a penalty factor $(\beta > 0)$, and $\beta = 10^{-4}$ for this paper; $\bar{L}$ is the operator

$$\bar{L} = \frac{1}{2} \begin{bmatrix} -\frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \tag{34}$$

The drilling rotations θ_z are linearly interpolated as

$$\theta_z = \sum_{i=1}^{3} L_i \theta_{zi} = \mathbf{F}_\theta \mathbf{q} \tag{35}$$

and the displacements $\boldsymbol{u}$ are linearly interpolated as

$$\mathbf{u} = \begin{Bmatrix} u \\ v \end{Bmatrix} = \begin{bmatrix} L_1 & 0 & L_2 & 0 & L_3 & 0 \\ 0 & L_1 & 0 & L_2 & 0 & L_3 \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{Bmatrix} = \mathbf{F}\mathbf{q} \tag{36}$$

Substituting Eqs. (35) and (36) into Eq. (33), the stiffness matrix of the element in the local coordinates is obtained as

$$\mathbf{K}^e = \mathbf{K}_m^e + \mathbf{K}_p^e + \mathbf{K}_\theta^e \tag{37}$$

The stiffness matrix $\mathbf{K}_\theta^e$ can be written as

$$\mathbf{K}_\theta^e = \beta \frac{Et}{(1-\nu^2)} \int_{V_e} \mathbf{B}_d^T \mathbf{B}_d \, \mathrm{d}V \tag{38}$$

where

$$\mathbf{B}_d = \mathbf{F}_\theta - \bar{L}\mathbf{F} \ . \tag{39}$$

In the global coordinate system (XYZ), for ease of transformation of the nodal displacement parameters of the flat shell element, $\theta_{y'} = -\theta_x$ will be used, and if $\theta_{x'} = \theta_y$, then the element stiffness matrix K^e becomes K^{*e}.

The transformation of the nodal displacements and rotations at node i are denoted by

$$\begin{Bmatrix} u^* \\ v^* \\ w^* \end{Bmatrix}_i = \mathbf{T}_i \begin{Bmatrix} U \\ V \\ W \end{Bmatrix}_i \quad \text{and} \quad \begin{Bmatrix} \theta_{x'} \\ \theta_{y'} \\ \theta_{z'} \end{Bmatrix}_i = \mathbf{T}_i \begin{Bmatrix} \theta_X \\ \theta_Y \\ \theta_Z \end{Bmatrix}_i \tag{40}$$

where $\mathbf{T}_i$ $(i = 1, 2, 3)$ is the transformation matrix.

Finally the element stiffness matrix $\bar{\mathbf{K}}^e$ for the nonconforming elements RDKTM15 or RDKTM18 (P) or RDKTM18 (T) in the global coordinates can be written as,

$$\bar{\mathbf{K}}^e = \bar{\mathbf{T}}^T \mathbf{K}^{*e} \bar{\mathbf{T}} \tag{41}$$

where

$$\bar{\mathbf{T}} = \begin{bmatrix} \mathbf{T}_1 & & & & & \\ & \mathbf{T}_1 & & & & \\ & & \mathbf{T}_2 & & & \\ & & & \mathbf{T}_2 & & \\ & & & & \mathbf{T}_3 & \\ & & & & & \mathbf{T}_3 \end{bmatrix} \tag{42}$$

According to Eq. (41) the vector of the nodal displacement parameters in the global coordinates $\mathbf{Q}^e$ can be expressed as

$$\mathbf{Q}^e = [U_1 \ V_1 \ W_1 \ \theta_{X1} \ \theta_{Y1} \ \theta_{Z1} \ U_2 \ V_2 \ W_2 \ \theta_{X2} \ \theta_{Y2} \ \theta_{Z2} \ U_3 \ V_3 \ W_3 \ \theta_{X3} \ \theta_{Y3} \ \theta_{Z3}]^T. \tag{43}$$

A full three-point quadrature is used to integrate the element stiffness matrix of the present elements RDKTM15, RDKTM18 (T) and RDKTM18 (P).

7 Numerical examples

In this section several existing triangular flat shell elements are selected for comparison with the proposed elements. These are

DKT15: the 15-DOF discrete Kirchhoff triangular flat shell element that is formed by the discrete Kirchhoff triangular plate element, DKT proposed by Batoz and Tahar [20], and the constant strain membrane element CST.

DKT18 (P): the 18-DOF discrete Kirchhoff triangular flat shell element that is formed by the DKT15 and triangular membrane elements proposed by Allman [13, 21].

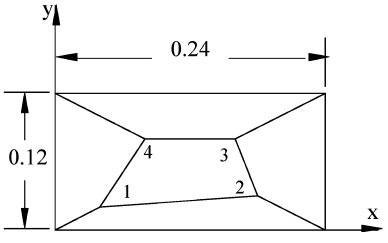
DKT18 (T) the 18-DOF discrete Kirchhoff triangular flat shell element that is formed by the DKT15 and triangular membrane elements with true drilling rotations.

The proposed elements are

RDKTM15: refined 15-DOF discrete triangular Mindlin flat shell element, in which Timoshenko's beam function is introduced, the re-constituting shear strain energy technique has been used, the stabilization stiffness matrix is added in order to suppress singularities of structural stiffness matrix.

RDKTM18 (P): refined 18-DOF discrete triangular Mindlin flat shell element, in which Timoshenko's beam function is introduced, the re-constituting shear strain energy technique has been used. The formulation of the membrane element is carried out with fictitious drilling rotations.

RDKTM18 (T): refined 18-DOF discrete triangular Mindlin flat shell element, in which Timoshenko's beam function is introduced, the re-constituting shear strain energy technique has been used. The formulation of the



node	coordinates	
	x_i	y_i
1	0.04	0.02
2	0.18	0.03
3	0.16	0.08
4	0.08	0.08

$E=10^5$ $\nu=0.25$ $t=0.01$

Fig. 2. Mesh for the patch test

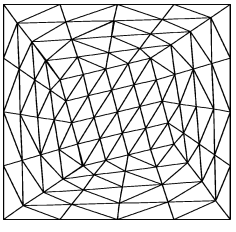
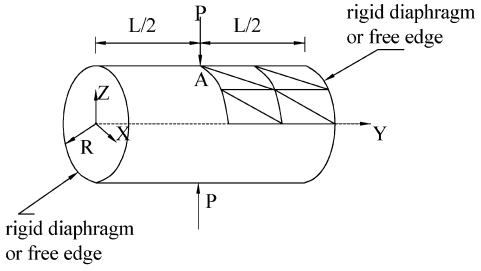


Fig. 3. A whole clamped plate divided by irregular meshes (160 elements)

Table 1. Central deflection w_c of a whole plate divided into an irregular mesh (160 elements) with different thickness/span ratios (clamped, uniform load)

h/L	0.6	0.4	0.25	0.2	0.05	10^{-2}	10^{-5}	10^{-7}	10^{-10}	10^{-15}	10^{-30}
DKT15	130.6	130.6	130.6	130.6	130.6	130.6	130.6	130.6	130.4	130.6	130.6
DKT18	130.6	130.6	130.6	130.6	130.6	130.6	130.6	130.6	130.4	130.6	130.6
RDKTM15	886.4	471.5	267.8	220.2	136.9	130.9	130.6	130.6	130.6	130.6	130.6
RDKTM18	886.4	471.5	267.8	220.2	136.9	130.9	130.6	130.6	130.6	130.6	130.6
EXACT							126.5	126.5	126.5	126.5	126.5



Diaphragm	Free edges (thickness=0.094)	Free edges(thickness=0.01546)
R = 300	R = 4.953	R=4.953
L = 600	L = 10.35	L=10.35
h = 3	h = 0.094	h=0.01546
$E = 3\times10^6$	$E = 10.5\times10^6$	$E=10.5\times10^6$
ν = 0.3	ν = 0.3125	ν = 0.3125
P = 100	P = 100	P = 0.025

Fig. 4. Pinched cylindrical shell with end diaphragms or with free edges (2 × 2 meshes)

membrane element is carried out with true drilling rotations.

Numerical examples

(1) Patch test for the plate bending and plane problem
The special cases of the patch test involving a plate bending patch test and a plane membrane patch test are used to start. The patch test for plate bending is adopted for a plate with an arbitrary mesh as shown in Fig. 2, and in which the boundary deflection is given to be $w = 1/2(1 + x + 2y + x^2 + xy + y^2)$ with $\theta_x = 1/2(2x + y)$, $\theta_y = 1/2(x + 2y)$. The numerical results show that the elements RDKTM15, RDKTM18, with full integration, all passed the aforementioned special patch test.

The patch test is presented to indicate that γ_x and γ_y can exactly express the zero constant shear strain and are free

Table 2. Central deflection (cylindrical shell with end diaphragms, central point load)

Element	2 × 2	4 × 4	6 × 6	8 × 8	10 × 10	12 × 12	24 × 24	36 × 36
DKT15	−0.000118	−0.000879	−0.001352	−0.001570	−0.001677	−0.001727	−0.001802	−0.001817
RDKTM15	−0.000118	−0.000894	−0.001353	−0.001573	−0.001680	−0.001733	−0.001813	−0.001830
DKT18(T)	−0.000118	−0.000898	−0.001355	−0.001572	−0.001677	−0.001726	−0.001818	−0.001833
DKT18(P)	−0.000128	−0.000954	−0.001388	−0.001582	−0.001677	−0.001721	−0.001793	−0.001807
RDKTM18(T)	−0.000118	−0.000899	−0.001358	−0.001577	−0.001686	−0.001736	−0.001818	−0.001833
RDKTM18(P)	−0.000128	−0.000955	−0.001391	−0.001589	−0.001686	−0.001740	−0.001809	−0.001825
EXACT			−0.001825					

Table 3. Central deflection (cylindrical shell with thickness = 0.094 and free edges, central point load)

Element	2 × 2	4 × 4	6 × 6	8 × 8	10 × 10	12 × 12
DKT15	−0.0961	−0.1075	−0.1107	−0.1119	−0.1124	−0.1128
RDKTM15	−0.0961	−0.1076	−0.1108	−0.1120	−0.1125	−0.1129
DKT18(T)	−0.0947	−0.1067	−0.1102	−0.1117	−0.1123	−0.1127
DKT18(P)	−0.0949	−0.1070	−0.1105	−0.1119	−0.1125	−0.1129
RDKTM18(T)	−0.0959	−0.1074	−0.1106	−0.1118	−0.1123	−0.1127
RDKTM18(P)	−0.0949	−0.1071	−0.1106	−0.1121	−0.1121	−0.1127
EXACT			−0.1139			

Table 4. Central deflection (cylindrical shell with thickness = 0.01546 and free edges, central point load)

Element	2 × 2	4 × 4	6 × 6	8 × 8	10 × 10	12 × 12
DKT15	−0.02152	−0.02366	−0.02417	−0.02437	−0.02446	−0.02454
RDKTM15	−0.02152	−0.02366	−0.02417	−0.02438	−0.02446	−0.02454
DKT18(T)	−0.02133	−0.02363	−0.02416	−0.02437	−0.02445	−0.02454
DKT18(P)	−0.02123	−0.02349	−0.02406	−0.02430	−0.02439	−0.02452
RDKTM18(T)	−0.02134	−0.02363	−0.02416	−0.02437	−0.02445	−0.02454
RDKTM18(P)	−0.02123	−0.02349	−0.02406	−0.02431	−0.02439	−0.02454
EXACT			−0.02439			

of shear locking. It can be noted that further numerical tests show that γ_x and γ_y can not pass the nonzero constant shear strain test, such as when the boundary deflection is given to be $w = x + 2y$ and $\theta_x = -1$, $\theta_y = -2$.

(2) A test for shear locking phenomena

A uniformly loaded whole clamped plate that is divided into irregular meshes (160 elements) with various ranges of thickness/span ratios as shown in Fig. 3 is used to test shear-locking phenomena. The results, presented in Table 1, indicate that the present element, RDKTM, shows the characteristics of being free from locking. Also it is shown that when the plate thickness is very small, the results of the present element are the same as those of the thin plate element DKT.

(3) Cylindrical shell with end diaphragms and with free edges

A cylindrical shell under a point load and with various types of end supports and thickness is used to demonstrate the accuracy of the present elements. This particular problem has been analyzed by many authors [22–24], and its geometry and other parameters are shown in Fig. 4. The analytical solution, which gives $w_A = -0.001825$ for a cylinder with end diaphragms, $w_A = -0.1139$ for a cylinder with thickness $= 0.094$ and free edges, and $w_A = -0.02439$ for a thin cylinder with thickness $= 0.01546$ and free edges, are compared with the numerical results in Tables 2–4. Note that the numerical results of the proposed elements are able to converge to the analytical solution, and that the results of the refined shell elements are always better than those of the other elements. It should be noted that the meshes used here are different from the special rectangular meshes in which a rectangle is divided into four triangular elements, an approach that is used by a number of researchers [22].

(4) Spherical shell

A spherical shell under two opposite radial concentrated loads at points A and B is shown in Fig. 5. The radial deflections at point A for different mesh divisions are used to compare with the analytical solution ($u_A = 0.094$ [22]). This example is often used for testing the properties of the poor response for membrane action, which is caused by using full integration for formulating the stiffness of the 18-DOF triangular flat shell element [16]. The results are compared in Table 5 and Fig. 7. It can be seen that the 15-DOF elements and 18-DOF elements with constraint of the true drilling rotation freedom produce the best results.

(5) Twisted beam problem

The twisted beam problem, shown in Fig. 6, has often been used to test the properties of the poor response for membrane action, which is caused by using full integration for formulating the stiffness of the 18-DOF triangular flat shell element when the thickness is very small [22]. The deflections along the loading direction at point C of the twisted beam with thickness of 0.0032 and 0.32 are given in Tables 6a and 6b, and the results of the twisted beam with thickness 0.0032 are shown in Fig. 8 for load case A and Fig. 9 for load case B. The numerical results show that the proposed elements are able to converge to the analytical solution and that the 15-DOF element RDKTM15 and 18-DOF element RDKTM18(T) with the constraint of true drilling rotation freedom produce the best results. Note that the results of the 18-DOF elements RDKTM18 (P) and

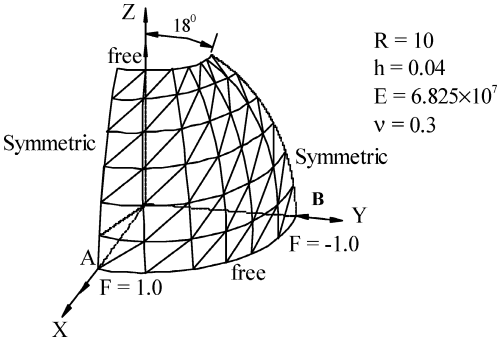


Fig. 5. A quarter of a hemispherical shell with an 18° hole on the top

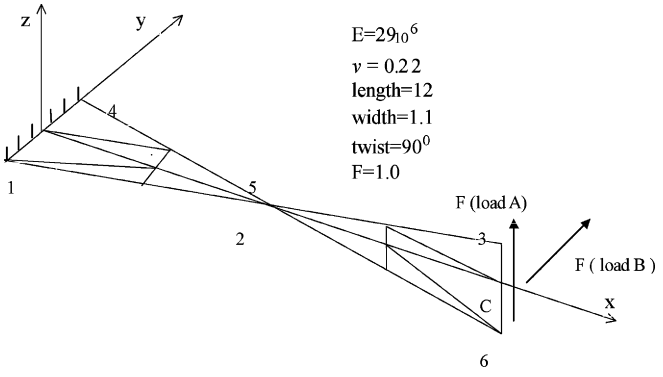


Fig. 6. Twisted beam divided into 2 × 12 mesh

Table 5. Radial deflections at the point A of spherical shell under concentrated loads at points A and B

Element	2 × 2	4 × 4	6 × 6	8 × 8	10 × 10	12 × 12	14 × 14	24 × 24	32 × 32
DKT15	0.08889	0.09384	0.093427	0.09353	0.09310	0.09280	0.09262	0.09260	0.09282
RDKTM15	0.08889	0.09384	0.09354	0.09311	0.09281	0.09262	0.09252	0.09261	0.09283
DKT18(T)	0.08627	0.09328	0.09333	0.09300	0.09274	0.09257	0.09248	0.09260	0.09283
DKT18(P)	0.04764	0.01910	0.02115	0.03541	0.05244	0.06635	0.07584	0.09064	0.09246
RDKTM18(T)	0.08627	0.09328	0.09333	0.09301	0.09275	0.09258	0.09249	0.09260	0.09283
RDKTM18(P)	0.04765	0.01911	0.02116	0.03542	0.05245	0.06636	0.07586	0.09247	0.09258
EXACT	0.094								

Table 6. Deflections at the point C of twisted beam

Element	Load A			Load B		
	2 × 8	2 × 12	2 × 16	2 × 8	2 × 12	2 × 16
a thickness = 0.0032						
DKT15	5256	5268	5274	1265.2	1284.4	1291.1
RDKTM15	5256	5268	5374	1265.2	1284.4	1291.1
DKT18(T)	4815	5174	5242	1170.3	1262.1	1282.8
DKT18(P)	81.949	185.34	389.74	20.301	46.043	96.489
RDKTM18(T)	4815	5174	5242	1170.3	1262.1	1282.8
RDKTM18(P)	81.950	185.39	389.82	20.302	46.043	96.489
EXACT		5256			1294	
b thickness = 0.32						
RDKTM15	0.005323	0.005396	0.005479	0.001393	0.001484	0.001547
RDKTM18(T)	0.005332	0.005399	0.005476	0.001397	0.001828	0.001845
RDKTM18(P)	0.005281	0.005366	0.005396	0.001629	0.001747	0.001758
EXACT		0.005424			0.001754	

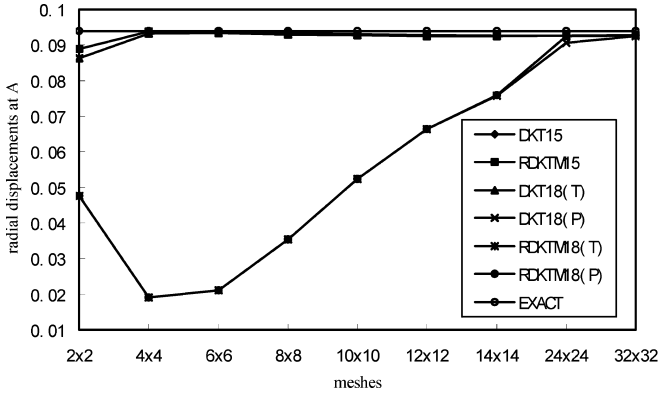


Fig. 7. Radial displacement at point A of spherical shell

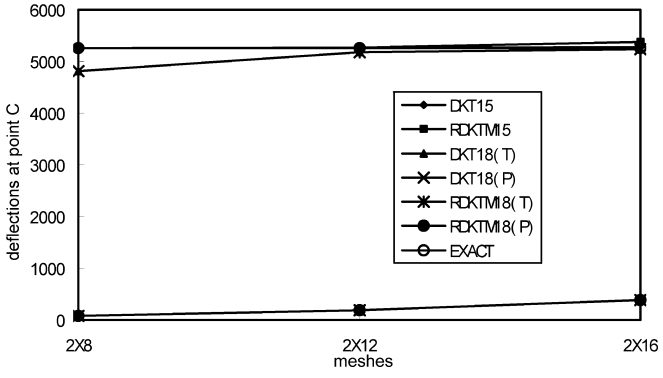


Fig. 8. Deflections at the point C of twisted beam with thickness = 0.0032 (load A)

RDKTM18 (T) with the constraint of drilling rotation freedom are better than those of the 15-DOF element RDKTM15 in case of the thickness 0.32.

8 Conclusions

The preceding determinations and numerical examples lead to several conclusions. First, The exact solution of Timoshenko's beam as a displacement of the boundary of the element can be used to derive the Mindlin flat shell

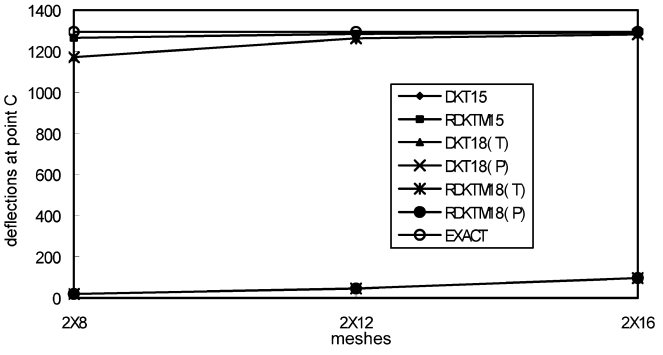


Fig. 9. Deflections at the point C of twisted beam with thickness = 0.0032 (load B)

element. This results in the thin and thick shell analysis being integrated in a uniform model and the convergence for very thin plate can be ensured theoretically. Second, the proposed discrete Mindlin flat shell element is a re-constitution shear strain element, which possesses high accuracy and has the ability for passing the patch test for Kirchhoff thin plate bending. It is also free from shear locking, and the Kirchhoff thin plate results can be obtained even for very thin plates with thickness/span ratios of 10^{-30}. This can be considered as the most efficient and simplest model among the existing ones. Third, In order to efficiently suppress singularities of the structural stiffness matrix, it is necessary to introduce true drilling rotations. Numerical examples are presented to show that the proposed element RDKTM18 (T) possesses high accuracy, especially for spherical shells and twisted beam structures. Finally, the formulations of the proposed elements RDKTM15 and RDKTM18 (T) are simple, and these elements can be used to solve thick/thin shell problems.

References

1. **Zienkiewicz OC, Taylor RL, Too JM** (1971) Reduced integration technique in general analysis of plates and shells. Int. J. Numer. Meth. Eng. 3: 275–290
2. **Pugh ED, Hiton E, Zienkiewicz OC** (1978) A study of triangular plate bending element with reduced integration. Int. J. Numer. Meth. Eng. 12: 1059–1078

3. **Malkus DS, Hughes TJR** (1978) Mixed finite element methods-reduced and selective integration techniques: a unification of concepts. Comput. Meth. Appl. Mech. Eng. 15: 63–81
4. **Hughes TJR, Cohen M, Haroun M** (1978) Reduced and selective integration techniques in finite element analysis of plates. Nucl. Eng. Design 46: 203–222
5. **Batoz JL, Lardeur P** (1989) A discrete shear triangular nine d.o.f. element for the analysis of thick to very thin plates. Int. J. Numer. Meth. Eng. 29: 533–560
6. **Batoz JL, Katili I** (1992) On a simple triangular Reissner/Mindlin plate element based on incompatible modes and discrete constraints. Int. J. Numer. Meth. Eng. 35: 1603–1632
7. **Katili I** (1993) A new discrete Kirchhoff-Mindlin element based on Mindlin-Reissner plate theory and assumed shear strain fields-part I: An extended DKT element for thick-plate bending analysis. Int. J. Numer. Meth. Eng. 36: 1859–1883
8. **Wanji Chen, Cheung YK** (2000) Refined quadrilateral element based on Midlin/Reissner plate theory. Int. J. Numer. Meth. Eng. 47: 605–627
9. **Wanji Chen, Cheung YK** (2001) Resined 9-DOF triangular Mindlin plate element. Int. J. Numer. Meth. Eng. 51: 1259–1281
10. **Zienkiewicz OC, Taylor RL** (1991) The Finite Element Method. 4th edn, Mcgraw-Hill Book Co, London
11. **Cook RD** (1993) Further development of a three-node triangular shell element. Int. J. Numer. Meth. Eng. 36: 1413–1425
12. **Irons BM, Ahmad** (1980) Techniques of Finite Elements. Ellis Horwood, Chichester
13. **Allman DJ** (1984) A compatible triangular element including vertex rotations for plane elasticity analysis. Comput. Struct. 19: 1–8
14. **Cook RD** (1994) Four-node 'flat' shell element: drilling degrees-of-freedom, membrane-bending coupling, warped geometry, and behavior. Comput. Struct. 50: 549–555
15. **Cook RD** (1987) A plane hybrid element with rotational D.O.F. and adjustable stiffness. Int. J. Numer. Meth. Eng. 24: 1499–1508
16. **Fish J, Belytschko T** (1992) Stabilized rapidly convergent 18 degrees-of-freedom flat shell triangular element. Int. J. Numer. Meth. Eng. 33: 149–162
17. **Wanji Chen, Cheung YK** (1999) Refined non-conforming triangular elements for analysis of shell structures. Int. J. Numer. Meth. Eng. 46: 433–455
18. **Providas E, Kattis MA** (2000) An assessment of two fundamental flat triangular shell elements with drilling rotations. Comput. Struct. 77: 129–139
19. **Sydenstricker RM, Landau L** (2000) A study of some triangular discrete Reissner-Mindlin plate and shell elements. Comput. Struct. 78: 21–33
20. **Batoz JL, Tahar MB** (1982) Evaluation of a new triangular thin plate bending element. Int. J. Numer. Meth. Eng. 18: 1655–1677
21. **Allman DJ** (1994) A basic flat facet finite element for the analysis of general shells. Int. J. Numer. Meth. Eng. 37: 19–35
22. **Sabourin F, Brunet M** (1995) Analysis of plate and shell with a simplified three node triangular element. Thin-Walled Struct. 21: 209–223
23. **Parisch H** (1979) A critical survey of the 9-node degenerated shell element with special emphasis in the thin shell application and reduced integration. Comput. Meth. Appl. Mech. Eng. 20: 323–350
24. **Jetteur P, Frey F** (1986) A four node Marguerre element for non-linear shell analysis. Eng. Comput. 3: 276–282